

Description

My notes about group theory. References below:

- 'Abstract algebra theory and applications, by T. Judson. <https://github.com/twjudson/aata>
- A book of abstract algebra, by C. Pinter: <https://math.umd.edu/~jcohen/402/Pinter%20Algebra.pdf>

Algebraic Structures

Question 1:

What is a relation R ?

Question 2:

What are operations?

Question 3:

What are the group properties of a group G ?

Question 4:

What is an abelian group?

Question 5: What is a ring?

Question 6: Examples of rings?

[Judson Pg 245]

Example:

Claim: the set of 2×2 matrices with entries of $\mathbb{R}$ is a ring under addition and multiplication.

Proof: Let $S = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mid a, b, c, d \in \mathbb{R} \right\}$. Then we must show S is a ring under addition and multiplication. This means that S is an abelian group with operation addition and also endowed with the binary operation of multiplication that satisfies distributivity and associativity.

First we show S is an abelian group under addition. Observe that

1. Identity: 0 is the element for the addition operation; this is a trivial fact.

2. Associativity: Clearly for any $U, V, T \in S$ we have $U + (V + T) = (U + V) + T$
3. Inverse: Finally, it is trivially true that any $U \in S$ has an inverse $U^{-1} \in S$ since it is just the additive inverse of U .
4. Commutativity: For any $U, V \in S$, it is trivial to check that $U + V = V + U$

Therefore, indeed S is an abelian group under addition. Now we must show that S is closed under the multiplication operation and satisfies the invariant properties of distributivity and associativity. I leave such a proof to future me, for it is a tedious thing to type.

Question 7: What are the types of rings?

Ring with Unity/Identity: A ring S is said to be a ring with identity if there is an identity element in S , call it e_2 , such that e_2 is an identity for the multiplication operation: $e_2 R_2 a = e_2 R_2 a = a$. Note I used the subscript 2 since e_1 is associated with the identity of the abelian group operation that S has (the addition operation)

Example

1. Set of binary strings under the $\vee$ and $\wedge$ operators

Commutative Ring: A ring is said to be a *commutative ring* if the R_2 binary operation is endowed with the commutativity invariant. That is, for $a, b \in S$ we have $a R_2 b = b R_2 a$

Examples of commutative rings:

1. Stabilizer group: $Stab(\psi)$. The stabilizer group is the set of stabilizer matrices and by definition they are commutative under multiplication, e.g. $ZI = IZ$

Question 8: What is a monoid?

Question 9:

Define polynomial

Question 10: Define polynomial $A[x]$

Question 11: Prove the following

Claim: Let A be a commutative ring with unity. We claim $A[x]$ is a commutative ring.

Question 12: What is a homomorphic map?

Question 13:

Example 1 (Judson Pg169): Let G be a group and $g \in G$. Define a map $L : \mathbf{Z} \rightarrow G$ by $L(m) = g^m$ where $m \in \mathbf{Z}$. Show that L is a homomorphic map from group $\mathbf{Z}$ to group G .